# AMIT SHARMA

Website, Github, Linkedin

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## PROFILE

- Background in Systems Science (Data Science ad Statistics)
- Strong skills in Python(4 years), R, SQL(2 years), Tableau(1 year). Familiar with Apache spark, Hadoop, Databricks, Azure, Parquet files.
- Strong understanding of Data mining, regression, clustering, neural network, predictive modelling.
- Excellent communication skills, problem solver, team player, multi-tasking. Eager to work in cross-functional team. Willing to relocate.

## SKILLS

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|---|---|
| Data Analysis | SQL , Spark, Python: pandas, matplotlib, numpy, gensim, scikit-learn (sklearn), pytorch (torch.nn), tensorflow, keras, Hadoop |
| Programming | Python, R, Tableau, C, Java, MATLAB, HTML5, CSS, REST API, NoSQL |
| Framework | Django, Scrapy, Jupyter |
| Other tools | Datbricks, , Git, Azure HDInsight, Databricks, MS Office |

## EXPERIENCE

**KataniTech** (March 2020 - Present)

**Role:** Artificial Intelligence and Backend Lead (Volunteer Position)
**Technology:** Django, Python, keras, HTML, CSS, Javascript, MongoDB

- Designed structure for the data collection, organization and ingestion for web-app.
- Selected methodology for data collection from social media to understand trends.
- Compiled data. Working on selection of algorithm for textile pattern generation and environment management for backend.
- Communicate the development of team with founder, create task assignment for team members (including myself) and document the ideas and methodology thought of and adopted.

**University of Ottawa (**September 2018 – December 2019)

**Project 1:** **Automatic Identification of Inventive Concepts of Patent Documents** (Bell Advanced Research Lab, Ottawa)
**Technology/Environment:** Python, Google Colaboratory, Jupyter
**Supervisor:** Professor Tet Hin Yeap
**Duration:** June 2019 – December 2019

- Defined the procedure for solving problem. Created data ingestion methods for using LDA, NNMF algorithm.
- Used pandas, sklearn, nltk, gensim, spacy and pyldavis for feature engineering and validating the model to decide which algorithm to adopt. Used pyLDAvis for visualizing the topic models.
- Identified major semiconductor businesses and their market competitions while carrying out backward citation analysis of documents.
- Automated data extraction from xml files of size over 1000MB. Data was extracted with web scrapping but result were polluted due to inconsistent result from search queries. Used xml library with assumption that quadratic blowup would not occur since USPTO is creating bulk data.

**Project 2:** **User Privacy in Data Mining**
**Technology/Environment:** Python, R
**Duration:** September 2019 – December 2019

- Synthesized dataset for demonstrating why Anonymization is not a very good method and worked on connecting Gaussian and Laplacian distribution for data randomization on query.
- Demonstrated how secure-multiparty computation or, randomization will help preserve privacy while allowing limited inference from a dataset, using the technique of Differential Privacy.

### Logistics Management Division, DOHS, Kathmandu, Nepal – July 2016 to June 2018

**Role:** Preventive Maintenance Engineer

**Responsibilities:**

- Prepared cost estimates of vehicle repair using current market valuation of the parts for The Secretary of Ministry of Health and Directors and Auditors at Department of Health Services.
- Prepared specification and cost estimates of vehicle to be purchased through competitive bidding.
- Completed size requirement analysis and specification design from scratch for competitive bidding of Refrigeration Truck to transport vaccine.

## EDUCATION

**Master of System Science** (Sept 2018 – Dec 2019), *University of Ottawa – Department of Computer Science & Electrical Engineering, Ottawa, CA*

**Bachelor of Engineering** (Nov 2011 – Oct 2015)**,** *Tribhuwan University, Lalitpur, Nepal*

**Online Courses**

6.86x: Machine Learning with Python-From Linear Models to Deep Learning
BMC: Bloomberg Market Concepts
*Microeconomics: The Power of Markets* by University of Pennsylvania on coursera
Game theory- Advanced Application, a course from Stanford University, on Coursera

## ACTIVITIES

**Professional Development Club, Founder and Webmaster,** University of Ottawa, Ottawa, CA
**System Science Graduate Student Association, Founder and President,** University of Ottawa, ON, CA

## HOBBIES

Hiking, Jogging, High intensity short duration workouts, Pushups

References:

Professor Tet Hin Yeap
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Professor Liam Peyton
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